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Effects of charge asymmetry on the liquid–liquid phase separation of polyampholytes and their condensate properties†

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Liquid–liquid phase separation (LLPS) is the mechanism underlying the formation of bio-molecular condensates which are important compartments regulating intra- and extra-cellular functions. Electrostatic interactions are some of the important driving forces of the LLPS behaviors of biomolecules. However, the understanding of the electrostatic interactions is still limited, especially in the mixtures of biomolecules with different charge patterns. Here, we focus on the electrostatic interactions in mixtures of charge-asymmetric and charge-symmetric polyampholytes and their roles in the phase separation behaviors. We build charge-asymmetric and charge-symmetric model proteins consisting of both glutamic acid (E, negatively charged) and lysine (K, positively charged), *i.e.* polyampholytes of E₃₅K₁₅ (charge asymmetric) and E₂₅K₂₅ (charge symmetric). Pure E₂₅K₂₅ can undergo LLPS. To investigate the effects of charge-asymmetric polyampholytes on the mixtures of E₂₅K₂₅/E₃₅K₁₅, we perform coarse-grained simulations to determine their phase separation. The charge-asymmetric polyampholyte E₃₅K₁₅ is resistant to the LLPS of the mixtures of E₂₅K₂₅/E₃₅K₁₅. The condensate density decreases with the molar fraction of E₃₅K₁₅ increasing to 0.4, and no LLPS occurs at the molar fraction of 0.5 and above. This can be attributed to the electrostatic repulsion between the negatively charged E₃₅K₁₅ polymers. We further investigate the effects of charge asymmetry on the conformations and properties of the condensates. The E₃₅K₁₅ polymers in the condensates exhibit a more collapsed state as the molar fraction of E₃₅K₁₅ increases. However, the conformation of E₂₅K₂₅ polymers changes slightly across different condensates. The surface tensions of condensates decline with the increase of the molar fraction of E₃₅K₁₅ polymers, while the diffusivity of polymers in the condensed phases is enhanced. This work elucidates the role of charge-asymmetric polyampholytes in determining the LLPS behaviours of binary mixtures of charge-symmetric and charge-asymmetric proteins as well as the properties of condensed phases.

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1. Introduction

Biomolecular condensates are being increasingly investigated for their critical roles in both intracellular and extracellular biological functions.^{1–4} In cells, they are also known as membraneless organelles. They are liquid-like organizations without enclosing membranes and are believed to form *via* the liquid–liquid phase separation (LLPS) of biomolecules. LLPS is driven by the overall intermolecular association, which includes electrostatic interactions, π – π /cation– π interactions, and hydrogen bonding, also known as multivalent interactions. Electrostatic

interactions have attracted a lot of research interest because many biomolecules, such as proteins/RNAs/DNAs, contain a substantial amount of charged groups.^{5–7} Intrinsically disordered proteins (IDPs) are a special class of proteins that do not adopt stable secondary or tertiary structures and are usually found in the biocondensates.⁵ They usually contain a large portion of negatively and/or positively charged amino acids, *i.e.* polyampholytes. One representative IDP is human prothymosin α (ProT α) that contains glutamic acid (E, negatively charged) and aspartic acid (D, negatively charged) residues in the central region and lysine (K, positively charged) residues at the two ends.^{7–9} It is a highly negatively charged IDP and forms a complex with the highly positively charged linker histone H1.^{10,11}

Tuning the charge patterning can manipulate the LLPS behaviors of IDPs.^{6,12–14} The relationship between charge patterning and the LLPS propensity of the polyampholytes has been established by theoretical and experimental approaches.^{12,13,15–18}

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Das, S. *et al.* systematically constructed coarse-grained models of polyampholytes consisting of glutamic acids E and lysine K by varying the block length of E or K.¹³ As the block length was increased from 1 to 25, *i.e.*, from (EK)₂₅ to E₂₅K₂₅, the critical phase separation temperature was increased. A similar observation was reported in other coarse-grained models of EK polyampholytes.^{6,14} This can be attributed to the enhanced intermolecular attractions as the block length of charged residues increases. Amin *et al.* systematically investigated the dissociation constant of two EK-polyampholyte chains with different sequence charge decorations (SCDs).¹⁸ The dissociation constant becomes less when the SCD is more negative; a more negative SCD represents a longer block length of like-charge residues. The critical phase separation temperature T_c increases as the dissociation constant becomes less. In our previous work,¹⁹ we calculated the second virial coefficients of a thousand pairs of polypeptides to characterize the intermolecular interaction strengths, which serve as an indicator of their phase separation propensity. The intermolecular interaction also impacts the single-chain conformation, which is highly correlated with the phase separation of multiple chains. Lin *et al.* reported the strong correlation between T_c and the radius of gyration (R_g) of single chains of EK-polyampholytes using coarse-grained modeling.¹⁶ In addition to computational work, Shi *et al.* validated the computational results on the effects of charge patterning by synthesizing the EK polyampholytes in experiments.¹² Phase separation was observed in E₁₆K₁₆ polyampholytes, while no LLPS occurred in (E₂K₂)₈. Most of these previous works were focused on the phase separation of charge-symmetric polyampholytes with a zero net charge. However, investigation of the charge-asymmetric polyampholytes with a nonzero net charge is still limited.^{20–22}

The charge asymmetry significantly impacts the structures and properties of the condensed phases. Compared with the charge-symmetric complex of polyelectrolytes, the charge-asymmetric complex exhibits lower glass transition temperature and moduli due to the plasticizing effect caused by increased water content.²¹ Mao *et al.* reported globule-to-coil transitions of protamine sequences, which are charge-asymmetric proteins abundant in arginine (R) residues but deficient in aspartic acid and glutamic acid,⁵ as the net charge of protamine increases. Danielsen *et al.* found little difference in the condensate density of the charge-asymmetric polyampholyte A₇₅^{−2/3}C₂₅⁺² (with a zero net charge) and charge-symmetric polyampholyte A₅₀^{−1}C₅₀⁺¹.²² A and C represent the negatively and positively charged blocks, the subscripts are the number of blocks, and the superscripts represent the block-charge valency. However, a systematic study is demanded to elucidate the role of charge-asymmetric polyampholytes in governing the phase separation and macroscopic properties of condensed phases.

In this study, we perform coarse-grained simulations to investigate the phase separation of mixtures of any two of the three representative polyampholytes, E₂₅K₂₅ (charge symmetric), E₃₅K₁₅ (charge-asymmetric), and E₁₅K₃₅ (charge-asymmetric). E represents glutamic acid with a negative charge, $-q$, and K is lysine with a positive charge, $+q$, as adopted from the coarse-grained

models in ref. 23. To understand the molecular mechanism underlying the phase behaviors of the mixtures, the intermolecular interactions are measured. Moreover, we investigate the effects of the charge asymmetry on the structure and properties of the condensed phases. The conformations of the polyampholytes in different condensates are characterized. The electrostatic, physical and dynamical properties of the obtained condensates are calculated. This work will advance the understanding of phase separation of charge-symmetric and charge-asymmetric polyampholytes, along with the structures and properties of condensates formed by different compositions of polyampholytes.

2. Methods

We perform direct-coexistence simulations to investigate the phase separation behaviors of polyampholytes, as adopted from ref. 19, 23 and 24. Coarse-grained simulations are computationally efficient and have been widely employed in the studies of macromolecular condensates.^{23,25}

2.1. Model description

The coarse-grained models of three polyampholytes, E₂₅K₂₅, E₃₅K₁₅, and E₁₅K₃₅, are constructed by using the IDP models developed in ref. 23. The coarse-grained models have demonstrated good agreement with experimental results in predicting the phase separation of disordered proteins as well as the macroscopic properties of condensed phases.^{26–28} Each amino acid is represented by one coarse-grained bead in implicit solvent. The coarse-grained beads are connected by harmonic bonds. Non-bonded potentials are represented by van der Waals (vdW) and coulombic (coul) interactions.

$$E_{\text{total}} = \sum_{i,j}^{N-1} k_{\text{bond}}(r_{i,i+1} - r_0)^2 + \sum_{i,j} \phi_{ij}^{\text{vdW}} + \sum_{i,j} \phi_{ij}^{\text{coul}} \quad (1)$$

where $r_{i,i+1}$ is the distance between two bonded beads i and $i + 1$. The harmonic bond force constant is 10 kcal mol^{−1} Å^{−2} and the equilibrium bond length, r_0 , is 3.8 Å. The short-range van der Waals interaction is represented by the Ashbaugh–Hatch potential.

$$\phi_{ij}^{\text{vdW}}(r) = \begin{cases} \phi_{\text{LJ}}(r_{ij}) + (1 - \lambda_{ij})\varepsilon & \text{if } r < 2^{\frac{1}{6}}\sigma_{ij} \\ \lambda_{ij}\phi_{\text{LJ}}(r_{ij}) & \text{if } r \geq 2^{\frac{1}{6}}\sigma_{ij} \end{cases} \quad (2)$$

where ϕ_{LJ} is the Lennard-Jones potential, $\phi_{\text{LJ}}(r_{ij}) = 4\varepsilon[(\sigma_{ij}/r_{ij})^{12} - (\sigma_{ij}/r_{ij})^6]$ with $\varepsilon = 0.2$ kcal mol^{−1}, r_{ij} is the distance between nonbonded residues i and j , and $\lambda_{ij} = (\lambda_i + \lambda_j)/2$ is used to describe the interaction strength between two residues, i and j , where λ_i , and λ_j are the hydrophobicity scaling factors of the amino acids i and j . Similarly, the distance $\sigma_{ij} = (\sigma_i + \sigma_j)/2$.

The long-range coulombic interactions between charged residues are described by a simple Debye–Hückel potential. Note, to keep the system charge neutral, counterions of lysine K are added to the E₂₅K₂₅/E₃₅K₁₅ systems (see Table 1). The simple Debye–Hückel potential may be limited in capturing the counterion–polymer interactions considering the

Table 1 The number of polymer chains and counterions in each system

Systems	Number of polymers/counterions			
	E ₂₅ K ₂₅	E ₃₅ K ₁₅	E ₁₅ K ₃₅	Counterion K
Pure E ₂₅ K ₂₅	500	—	—	—
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.9/0.1)	450	50	—	1000
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.8/0.2)	400	100	—	2000
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.7/0.3)	350	150	—	3000
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.6/0.4)	300	200	—	4000
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.5/0.5)	250	250	—	5000
E ₂₅ K ₂₅ /E ₃₅ K ₁₅ (0.3/0.7)	150	350	—	7000
E ₃₅ K ₁₅ /E ₁₅ K ₃₅ (0.5/0.5)	—	250	250	—

complicated and diverse effects of counterions on the LLPS of polyampholytes/polyelectrolytes.^{29,30} We remind readers of the limitations of the models in the presence of counterions in this study.

$$\phi_{ij}^{\text{coul}}(r) = \frac{q_i q_j}{4\pi D r} e^{-\kappa r} \quad (3)$$

where q_i is the charge of residue i located at the center of the corresponding bead, $D = 80$ is the dielectric constant of the solvent medium (water), and κ is the inverse Debye screening length of 1 nm^{-1} , corresponding to an monovalent ion concentration of approximately 100 mM.

2.2. Simulation details

To determine the phase separation, we perform direct-coexistence simulations using a slab geometry as shown in Fig. 1. The number of polymer chains and counterions K is listed in Table 1. The total number of polymers in each system is fixed at 500, and the number of counterions K is determined to keep the whole system

charge neutral. We take the following steps to obtain the initial slab-geometric structures of the simulations, as described in ref. 31. (i) Polymers and counterions are randomly placed in a cubic box of $300 \times 300 \times 300 \text{ \AA}^3$. We performed a simulation in the NPT ensemble at 300 K and 5–10 bar. The temperature is kept constant using the Nosé–Hoover thermostat. The pressure is maintained by a Parrinello–Rahman barostat. The pressure of 5–10 bar is determined to obtain a compressed simulation box with a density of around $0.6\text{--}0.7 \text{ g mL}^{-1}$. (ii) Starting with the compressed structure, we extend the z-direction to 1600 Å and perform simulations in a canonical NVT ensemble at 300 K for at least 1 μs. A Langevin dynamics thermostat is applied with a damp factor of 1 ps^{-1} to keep the temperature constant. Periodic boundary conditions are applied in all the three directions. A time step of 10 fs was used for all the simulations, and the trajectory is recorded every 1000 time steps. Simulations were conducted using the LAMMPS package.³²

2.3. Property calculation

2.3.1. Potential of mean force. To measure the intermolecular interaction strength between a pair of polymers, we calculate the potential of mean force (PMF, also known as free energy A) of two polymer chains as a function of their center-of-mass distance (r_{COM}).¹⁹ The PMF is calculated using the adaptive biasing force (ABF) method.^{33,34} By using the ABF algorithm, an external biasing force is applied to encourage a more evenly distributed sampling of the distance. The negative of the gradient of the potential of mean forces is the average forces acting along the transition coordinate. Therefore, the potential of mean forces can be estimated simply by summing the

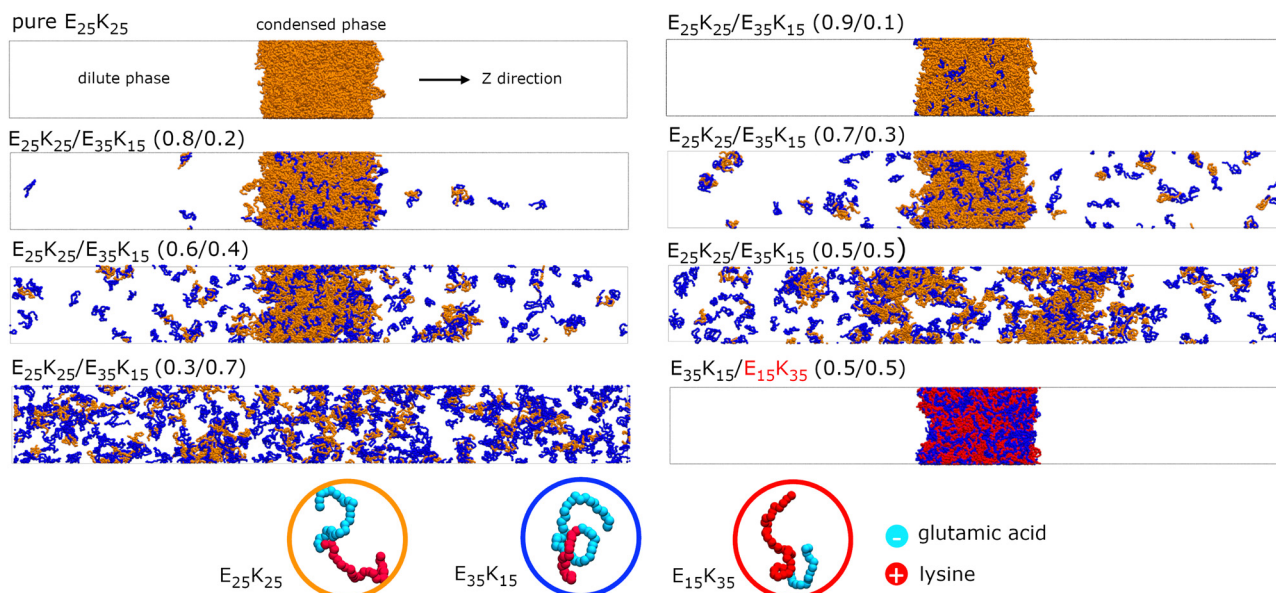


Fig. 1 The snapshots of slab-geometry direct-coexistence simulations of pure E₂₅K₂₅ polymers and the binary mixtures of E₂₅K₂₅/E₃₅K₁₅ and E₃₅K₁₅/E₁₅K₃₅ at 1 μs. Counterions are not shown for clarity. The snapshots of the three polyampholytes are shown at the bottom: the charge-symmetric E₂₅K₂₅ (orange circle), the charge-asymmetric E₃₅K₁₅ (blue circle), and the charge-asymmetric E₁₅K₃₅ (red circle). Glutamic acid and lysine are represented by red and cyan beads, respectively. The total number of polymers in each system is 500 and their molar ratios are shown in parentheses. The number of counterions K is listed in Table 1 to keep the whole system charge neutral. Direct coexistence simulations are performed in the canonical ensemble at 300 K.

average force in individual bins. The equations shown below are adopted from ref. 34.

$$\Delta A_{\xi} = - \sum_{k=1}^M \overline{F}_{\xi}(N_{\text{step},k}) \Delta \xi \quad (4)$$

where ξ is the transition coordinate, M is the number of bins, and $\overline{F}_{\xi}$ is the average force in the bin k .

$$\overline{F}(N_{\text{step}}, k) = \frac{1}{N_{\text{step}}^k} \sum_{\mu=1}^{N_{\text{step}}^k} F_{\mu,k} \quad (5)$$

where N_{step}^k is the number of samples accrued in bin k after N_{step} steps and $F_{\mu,k}$ is the μ th force sample in this bin.

In the simulations, two polymer chains are initially placed in a $300 \times 300 \times 300 \text{ \AA}^3$ box. The transition coordinate is the center-of-mass distance between two polymers, r_{COM} . To keep r_{COM} within $0\text{--}100 \text{ \AA}$, a harmonic force with a magnitude of $50.0 \text{ kcal mol}^{-1} \text{ \AA}^2$ is applied on the upper and lower boundaries of the window of $r_{\text{COM}} = 1\text{--}100 \text{ \AA}$. Force statistics is stored in bins of width $1 \text{ \AA}$. The biasing force is applied after 1000 samples are collected in each bin. A final production run of $1 \text{ \mu s}$ is performed. Three independent simulations are performed for each free energy calculation to collect the statistical deviations.

2.3.2. Surface tension. The surface tension γ of the condensed phase is determined in slab geometry by expanding the z -axis of a cubic cell by $900 \text{ \AA}$, adopted in ref. 35 and 36. The cubic cell is composed of fully equilibrated bulk simulation of 500 polymer chains and the counterions. The simulations are performed in the NVT ensemble. A timestep of 10 fs is used to integrate the equations of motion. Periodic boundary conditions are applied in the three directions. The production run is 100 ns . The pressure tensor (P_{xx} , P_{yy} , and P_{zz}) is printed every time step. The printed pressure at the last 60-ns of the simulation is used for calculating the surface tension.

$$\gamma = \frac{L_z}{2} \left\langle P_{zz} - \frac{P_{xx} + P_{yy}}{2} \right\rangle \quad (6)$$

where L_z is the box length in the z direction, P_{xx} , P_{yy} , P_{zz} are the diagonal elements of the pressure tensor and $\langle \rangle$ denotes the ensemble average.

To determine the average and mean values of the surface tension, we subsample the pressures in the last 60 ns to obtain statistically independent data and divide them into six blocks. Specifically, we first calculated the statistical inefficiency, g , of the pressures in the last 60 ns of the trajectory. Based on this, we sample N/g number of uncorrelated pressure data, where N is the total number of recorded pressure values in the last 60 ns of the trajectory. Subsampled uncorrelated pressure data are then divided into 6 independent blocks for calculating the average surface tension and its standard deviation. The pymbar package is used to calculate the statistical inefficiency and generate the subsampled uncorrelated data.³⁷

2.3.3. Self-diffusion coefficient. Einstein's relation can describe the self-diffusion coefficient (D) of a 3-dimensional system:^{38,39}

$$D = \lim_{\Delta\tau \rightarrow \infty} \frac{1}{6\Delta\tau} \langle |\vec{r}_{t+\Delta\tau} - \vec{r}_t|^2 \rangle, \quad (7)$$

where $\vec{r}$ is the position of the center-of-mass of a tagged chain, $\langle \rangle$ denotes the ensemble average, and $|\vec{r}_{t+\Delta\tau} - \vec{r}_t|^2$ is the mean squared displacement (MSD) of the center-of-mass of a polymer. The MSD is calculated by analyzing the last 90 ns of the $1\text{-}\mu\text{s}$ trajectory obtained in the direct-coexistence simulations. During the $910\text{--}1000 \text{ ns}$ of the trajectory, we visually observe in VMD (visual molecular dynamics⁴⁰) to ensure almost all the polymers in the condensate phase move within the condensate phase and not cross the condensate/dilute phase interface. The last 90-ns trajectory is divided into three blocks to calculate the mean and standard deviations.

3. Results and discussion

3.1. Charge asymmetry is unfavorable for the phase separation of binary mixtures of charge-symmetric and charge-asymmetric polyampholytes E_mK_n

To investigate the effects of charge-asymmetry on the phase separation of polyampholytes, we start with the mixtures of polyampholytes $E_{35}K_{15}$ (charge asymmetric, a net charge of $-20q$) with the charge-symmetric $E_{25}K_{25}$. It was reported previously that pure $E_{25}K_{25}$ polymers can be self-organized forming a condensed phase *via* the LLPS.^{13,14,41} We obtain consistent results by performing the direct-coexistent simulation of pure $E_{25}K_{25}$ (see Fig. 1). Based on this, $E_{35}K_{15}$ is added to the pure $E_{25}K_{25}$ system to investigate its effect on the phase separation of the binary mixtures. Counterions of K are utilized to keep the systems charge neutral. The runtime of the simulations is at least $1 \text{ }\mu\text{s}$. The density in the condensate phases (if LLPS occurs) is calculated every 100 ns to ensure equilibrium (see Fig. S1 in the ESI†).

The addition of the $E_{35}K_{15}$ polymers is unfavorable for the LLPS of the mixtures of $E_{25}K_{25}/E_{35}K_{15}$. The condensate density decreases from 0.54 g mL^{-1} to 0.36 g mL^{-1} with the molar fraction increasing from 0 to 0.4 (see Fig. 2). Here, the condensed phase is defined as the region where variations in the average density are within 10% of the maximum value.¹⁹ In the condensed phases, the standard deviations of the density increase significantly as the initial molar fraction of $E_{35}K_{15}$ increases. The interface between the coexistent condensed and dilute phases becomes unstable until no LLPS occurs at the molar fractions of 0.5 and 0.7 (Fig. 1). At the $E_{35}K_{15}$ molar fraction of 0.5, large aggregates are observed along with small clusters. The large aggregates disassemble as the molar fraction of $E_{35}K_{15}$ further increases to 0.7. The decreased LLPS propensity can be attributed to the electrostatic repulsion between negatively charged $E_{35}K_{15}$, which we will elaborate in Section 3.2.

The compositions in the condensed phases vary with the initial fraction of $E_{35}K_{15}$. We calculate the density of each component in the $E_{25}K_{25}/E_{35}K_{15}$ mixtures. In the $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1) system, the densities $E_{25}K_{25}$ and $E_{35}K_{15}$ in the

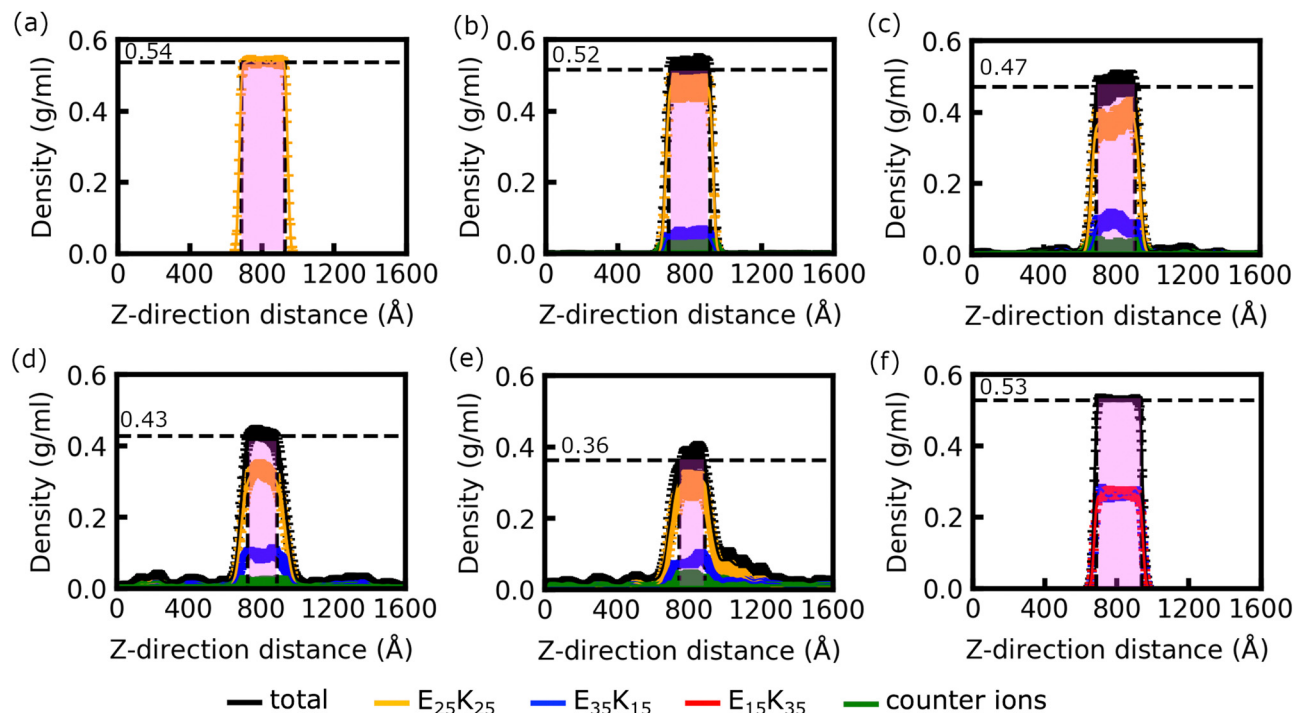


Fig. 2 The density profiles of (a) pure $E_{25}K_{25}$, (b) $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1), (c) $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2), (d) $E_{25}K_{25}/E_{35}K_{15}$ (0.7/0.3), (e) $E_{25}K_{25}/E_{35}K_{15}$ (0.6/0.4), and (f) $E_{35}K_{15}/E_{15}K_{35}$ (0.5/0.5) systems in the z-direction as shown in Fig. 1. Horizontal black dashed lines represent the average condensate density. Vertical black dashed lines are the boundaries of the condensate regions shaded in purple. Condensate regions are defined as the areas where the variations in the average density are within 10% of the maximum value across the simulation box. Simulations of 500 polymer chains in each system are performed in a canonical ensemble at 300 K for at least 1 μ s. The last 100-ns trajectories are analyzed to calculate the average density profiles. Error bars represent the standard deviations over the last 100-ns trajectories.

condensed phase are 0.468 g mL^{-1} and 0.052 g mL^{-1} , respectively. This corresponds to an $E_{25}K_{25}/E_{35}K_{15}$ molar ratio of around 0.9/0.1, which is the same as the initial value. Similarly, the molar ratio of $E_{25}K_{25}$ to $E_{35}K_{15}$ in the $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2) system is close to their initial ratios. However, in the $E_{25}K_{25}/E_{35}K_{15}$ (0.7/0.3 and 0.6/0.4) systems, the molar ratios of $E_{25}K_{25}$

to $E_{35}K_{15}$ in the condensed phases are 1/0.3 and 1.4/0.4, respectively, much higher than their corresponding initial values. This is because of the fact that a significant number of $E_{35}K_{15}$ molecules are observed in the coexistent dilute phase (see Fig. 3). Regardless of the difference in the compositions of the two components $E_{25}K_{25}$ and $E_{35}K_{15}$, they are uniformly

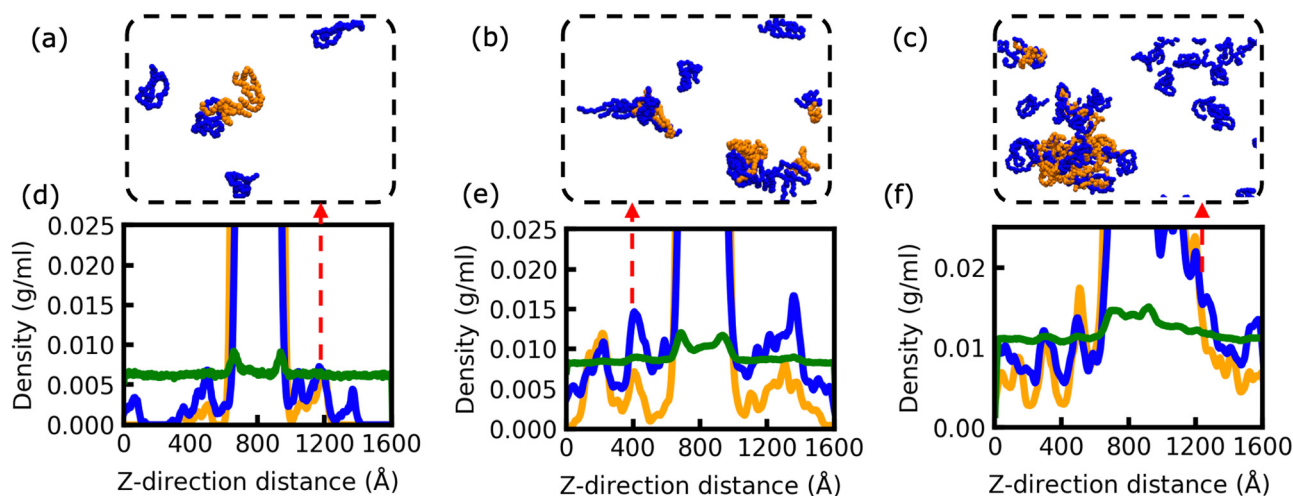


Fig. 3 The snapshots of dispersed $E_{35}K_{15}$ chains and their aggregates with $E_{25}K_{25}$ in the dilute phases of (a) $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2), (b) $E_{25}K_{25}/E_{35}K_{15}$ (0.7/0.3), and (c) $E_{25}K_{25}/E_{35}K_{15}$ (0.6/0.4) systems. (d, e, and f) The density profiles of the coexistent dilute phases. Counterions are not shown in (a)–(c) for clarity. The color scheme is the same as that in Fig. 1 and 2. Red dashed lines indicate the position of representative clusters in the dilute phases.

distributed in all the condensed phases. No segregates of these two polymers are observed, in contrast to subcompartments inside a condensate formed by two polyampholytes with significantly different charge patterns.⁴²

In the dilute phases of the $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2, 0.7/0.3 and 0.6/0.4) systems, both aggregates of $E_{25}K_{25}/E_{35}K_{15}$ and dispersed chains of $E_{35}K_{15}$ are observed (see Fig. 3a–c). At the $E_{35}K_{15}$ molar fraction of 0.2, the aggregates consist of two to four polymers, while they contain more than 10 polymer chains at the molar fraction of 0.4. Previous theoretical models assume that polymers in the dilute phase are either dispersed chains or aggregates.^{43,44} By assuming that the polymers exist as dispersed chains in the dilute phase, the theoretical models are limited in predicting the dilute-phase density.^{28,45–47} Zhang improved the theoretical models of polyelectrolytes by considering clusters in the supernatant phases of polyelectrolyte coacervates.⁴³ Veis *et al.* also suggested that aggregates of oppositely charged polyions exist in the supernatant phase of polyelectrolytes.⁴⁸ Complementary to the previous theoretical work, the coarse-grained modeling in this study proved the coexistence of both aggregates and dispersed chains in the dilute phases. The underlying mechanisms will be discussed in Section 3.2. Besides the mixtures of $E_{25}K_{25}/E_{35}K_{15}$, we intuitively mix the two oppositely charged polymers $E_{35}K_{15}$ and $E_{15}K_{35}$ with an equal molar fraction (see Fig. 1). Expectedly, they are associated forming a condensed phase due to the electrostatic attraction. The average density of the condensate phase is 0.53 g mL^{-1} , similar to those of the pure $E_{25}K_{25}$ and $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1) condensates (see Fig. 2).

3.2. Intermolecular repulsion and attraction underlie the phase separation of the mixtures of polyampholytes E_mK_n

To unveil the mechanisms of the phase behaviors of the polyampholyte mixtures described in Section 3.1, we calculate the intermolecular interactions between different pairs of polymers. We perform simulations to measure the potential of mean forces (PMFs) between two polymers using the adaptive biasing force (ABF) method.³³ The reaction coordinate is the distance between the centers of mass r_{COM} of two polymers.

The $E_{35}K_{15}$ – $E_{35}K_{15}$ pairs are repulsive. As the center-of-mass distance r_{COM} decreased from 100 Å to 1 Å, the PMF between the two $E_{35}K_{15}$ polymers gradually increases above zero (see Fig. 4a), indicating the increased repulsion between $E_{35}K_{15}$ polymers. It is noticeable that the repulsion does not monotonically increase, but with a bump at around 28 Å. Beyond 28 Å, the two polymers act like two negatively charged spherical particles. The average radius of gyration of a single $E_{35}K_{15}$ polymer, R_g , is 14.2 Å (see Fig. 4b), corresponding to the distance of 28.4 Å when the two polymeric particles collide. However, when the two $E_{35}K_{15}$ polymers approach below 28 Å, they are interpenetrated. The oppositely charged amino acids (E and K) in the two polymers attractively interact, leading to the interfusing of the two chains and thus a decrease in the repulsion of the two $E_{35}K_{15}$ polymers. The $E_{25}K_{25}$ – $E_{25}K_{25}$ and $E_{25}K_{25}$ – $E_{35}K_{15}$ pairs are attractive. The PMF between $E_{25}K_{25}$ and $E_{25}K_{25}$ decreases below zero as the polymer pair approaches

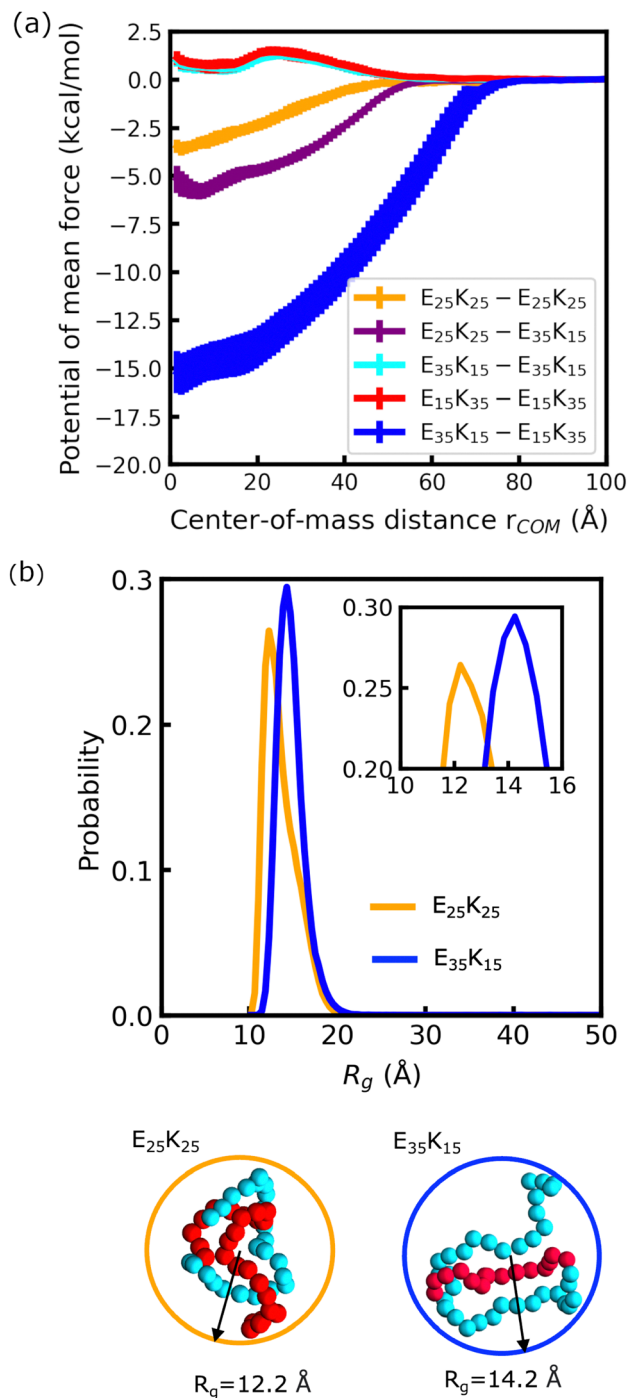


Fig. 4 (a) The potentials of mean force (PMFs) between the centers of mass of five pairs of polyampholytes at 300 K: $E_{25}K_{25}$ – $E_{25}K_{25}$, $E_{25}K_{25}$ – $E_{35}K_{15}$, $E_{35}K_{15}$ – $E_{35}K_{15}$, $E_{15}K_{35}$ – $E_{15}K_{35}$, and $E_{35}K_{15}$ – $E_{15}K_{35}$. Note that the PMF profiles of $E_{35}K_{15}$ – $E_{35}K_{15}$ and $E_{15}K_{35}$ – $E_{15}K_{35}$ are overlapped. Error bars represent the standard deviations by performing three replicate simulations. (b) The distribution of the radius of gyration R_g of a single polymer chain: $E_{25}K_{25}$ and $E_{35}K_{15}$, and the representative snapshots of their conformations. Cyan and red beads represent glutamic acid (E) and lysine (K), respectively.

each other, suggesting an increased intermolecular association. Similarly, the $E_{25}K_{25}$ – $E_{35}K_{15}$ pair exhibits intermolecular attraction. The attraction is stronger than that of the $E_{25}K_{25}$ – $E_{25}K_{25}$

pair, as evidenced by the more negative PMF values. We attribute this to the weak intramolecular interaction of the $E_{35}K_{15}$ polymer. The charge-asymmetric $E_{35}K_{15}$ exhibits a slightly elongated globule state due to the excess charge,⁴⁹ as shown by a higher average R_g value than that of $E_{25}K_{25}$ (see Fig. 4b). The excess charge in $E_{35}K_{15}$ reduces its intramolecular interaction, but enhances its interaction with the $E_{25}K_{25}$ polymer, leading to more negative PMF values than those between $E_{25}K_{25}$ and $E_{25}K_{25}$. On the other hand, with a symmetric charge patterning, the negatively and positively charged residues of $E_{25}K_{25}$ in one single chain can match resulting in a compactly collapsed state (see Fig. 4b). This reduces its interaction with the other polymer, thus weakening the intermolecular interaction between $E_{25}K_{25}$ polymers. We also determined the potential of mean force of the $E_{35}K_{15}$ - $E_{15}K_{35}$ pair, which is $-15 \text{ kcal mol}^{-1}$ at $r_{\text{COM}} = 1.0 \text{ \AA}$, the most negative value among all the pairs at the same r_{COM} .

The competition between the repulsion ($E_{35}K_{15}$ - $E_{35}K_{15}$) and the attraction ($E_{25}K_{25}$ - $E_{25}K_{25}$ and $E_{25}K_{25}$ - $E_{35}K_{15}$) determines the phase separation of mixtures of $E_{25}K_{25}$ / $E_{35}K_{15}$ as discussed in Section 3.1. As the molar fraction of $E_{35}K_{15}$ increases, the repulsion becomes stronger. The density of the condensed phases decreases (in Fig. 2), suggesting unstable condensed phases. When it is further increased to 0.5 (see Fig. 1), the repulsion overtakes the attraction and no LLPS is observed.

In addition to the condensate phase, the intermolecular repulsion and attraction affect the conformations of polymers in the dilute phase. Some $E_{35}K_{15}$ chains are dispersed due to the repulsion between $E_{35}K_{15}$ polymers. Meanwhile, the attraction between $E_{25}K_{25}$ and $E_{35}K_{15}$ polymers results in clusters of $E_{25}K_{25}$ and $E_{35}K_{15}$ chains in the dilute phases in Fig. 3a-c.

3.3. $E_{35}K_{15}$ demonstrates a conformational change across different condensates

To investigate the effects of charge asymmetry on the structures of polymers in the condensed phases, we calculate the R_g of $E_{25}K_{25}$ and $E_{35}K_{15}$ in different condensates as shown in Fig. 5.¹² The distribution of R_g of $E_{25}K_{25}$ is almost unchanged in the three condensates, pure $E_{25}K_{25}$, and $E_{25}K_{25}$ / $E_{35}K_{15}$ (0.9/0.1 and 0.8/0.2), and slightly shifts to the left as the molar fraction of $E_{35}K_{15}$ increases to 0.4 (see Fig. 5a). On the other hand, the $E_{35}K_{15}$ polymer exhibits significantly different conformations in $E_{25}K_{25}$ / $E_{35}K_{15}$ (0.7/0.3 and 0.6/0.4) condensates from that in $E_{25}K_{25}$ / $E_{35}K_{15}$ (0.9/0.1 and 0.8/0.2) condensates (see Fig. 5b). In the $E_{25}K_{25}$ / $E_{35}K_{15}$ (0.9/0.1 and 0.8/0.2) condensates, they expand to a coil-like state with the peak at $21.2 \text{ \AA}$, in contrast to the peak at $14.2 \text{ \AA}$ for the $E_{25}K_{25}$ / $E_{35}K_{15}$ (0.7/0.3 and 0.6/0.4) condensates.

The molar fraction of $E_{35}K_{15}$ impacts the inter- and intra-molecular interactions, thus leading to the conformation

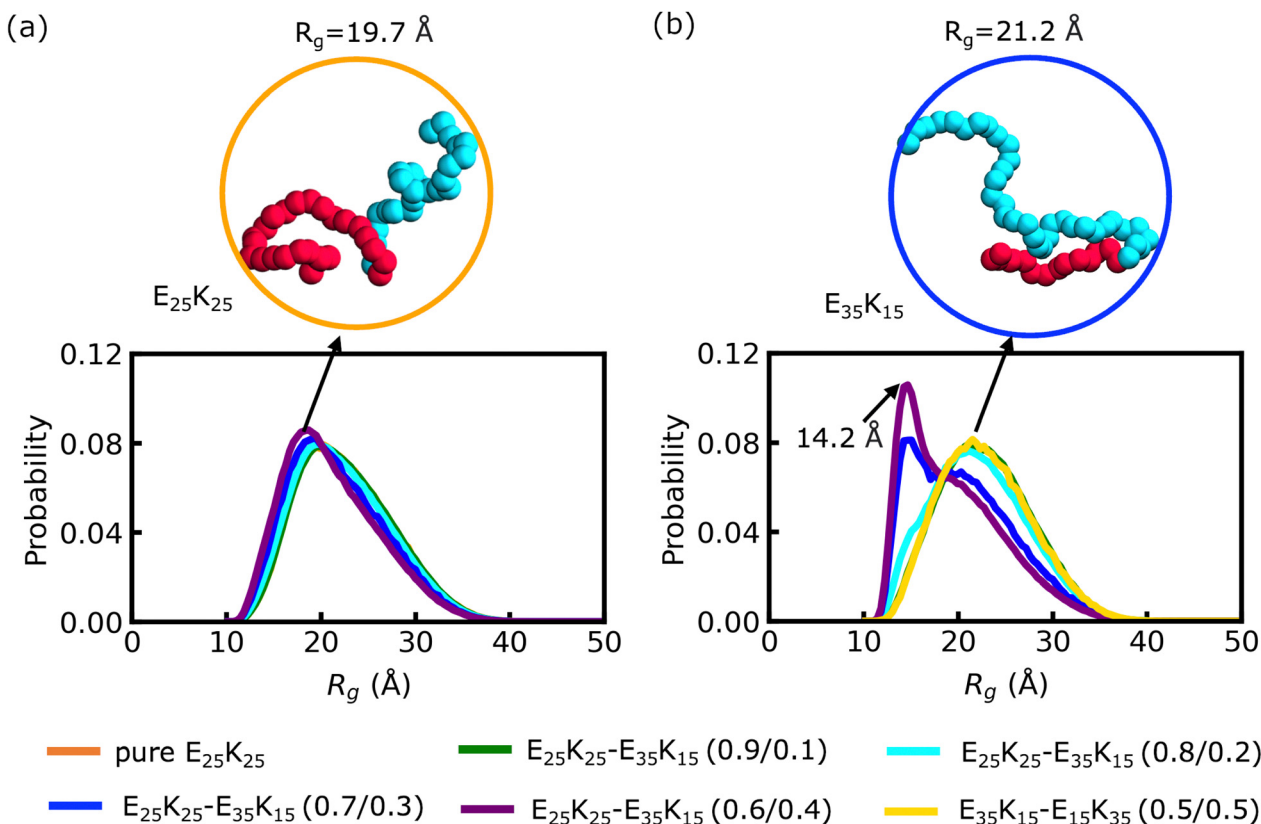


Fig. 5 The distributions of the radius of gyration (R_g) of (a) $E_{25}K_{25}$ and (b) $E_{35}K_{15}$ in different condensates. In (a), the orange and green lines are overlapped with the cyan line, and in (b) the green line is overlapped by the gold line. Snapshots of the representative conformations of $E_{25}K_{25}$ and $E_{35}K_{15}$ in the condensed phases are shown as the orange and blue circles, respectively.

change of $E_{35}K_{15}$ in the condensed phases. At a low molar fraction of $E_{35}K_{15}$, 0.1 and 0.2, all of the $E_{35}K_{15}$ polymers can interact attractively with $E_{25}K_{25}$ molecules and exhibit a coil-like state. However, as the molar fraction increases, the overall intermolecular association is significantly diminished due to the increased intermolecular repulsion between $E_{35}K_{15}$ polymers. Only a portion of $E_{35}K_{15}$ in the condensates can be associated with $E_{25}K_{25}$ polymers exhibiting a coil-like state, while the others collapsed and behave more like dispersed chains in the condensates (see Fig. 4b). The peak position of the R_g distribution of $E_{35}K_{15}$ in Fig. 5b is at 14.2 Å, similar to that of a dispersed chain shown in Fig. 4b. The dispersed $E_{35}K_{15}$ polymers are in a globule state due to intramolecular interactions, similar to the snapshot shown in Fig. 4b.

Although $E_{35}K_{15}$ polymers tend to transform to a globule state with increasing molar fractions of $E_{35}K_{15}$, the average R_g values of $E_{35}K_{15}$ in the condensates ($E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1 to 0.6/0.4) (Fig. 5b) are still higher than that of a single chain (Fig. 4b). The average R_g values of $E_{35}K_{15}$ in the condensates ($E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1 to 0.6/0.4) are 19.6 to 22.9 Å, while that of a single polymer chain is 14.7 Å. Similarly, the average R_g of $E_{25}K_{25}$ in the condensate is larger than that of a single $E_{25}K_{25}$ chain, where the former is 22.5 Å, and the latter is 13.6 Å.

We also compare the R_g distribution of polyampholytes in the condensate phase with that in the dilute phase (see Fig. S2 of the ESI†). Note that in the three systems, pure $E_{25}K_{25}$, $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1), and $E_{35}K_{15}/E_{15}K_{35}$ (0.5/0.5), no polymers are observed in the dilute phase. We only analyze the other three systems $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2, 0.7/0.3, and 0.6/0.4). In all these three systems, the peak of the R_g distribution in the dilute phase is located left to that in the condensate phases and the peak intensity is higher, indicating the polymers tend to be in a globule state in the dilute phase, but a coil-like state in the condensate phase. As the molar fraction of $E_{35}K_{15}$ increases from 0.2 to 0.4, the difference in the R_g distributions between the condensate and dilute phases, *e.g.* peak position and intensity, is reduced.

3.4. Condensates of $E_{25}K_{25}/E_{35}K_{15}$ are negatively charged with low surface tensions and high diffusivity

In addition to the structural properties, macroscopic properties of condensates are crucial to their biological applications.^{28,50–52} We focus on the electrostatic, interfacial, and dynamical properties. The electrostatic properties of condensates govern their binding with other biomolecules, *e.g.* RNAs, and emulsion stability.^{50,53} The pure $E_{25}K_{25}$ and $E_{35}K_{15}/E_{15}K_{35}$ (0.5/0.5) condensates are both charge neutral, while those of $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1, 0.8/0.2, 0.7/0.3, and 0.6/0.4) are negatively charged including both polymers and counterions (see Fig. 6). As the molar fraction of $E_{35}K_{15}$ increases from 0.1 to 0.2, the average charge density of the condensate increases from 0.08 to 0.10 q/nm^3 (negatively charged) and changes slightly as the molar fraction further increases to 0.4.

Surface tension plays an important role in the coalescence of condensed phases.^{50,54,55} To account for the effects of $E_{35}K_{15}$ on the surface tension γ , we normalize it with respect to that of the pure $E_{25}K_{25}$ condensate γ_0 , *i.e.* $\gamma^* = \gamma/\gamma_0$.²⁸ The normalized surface

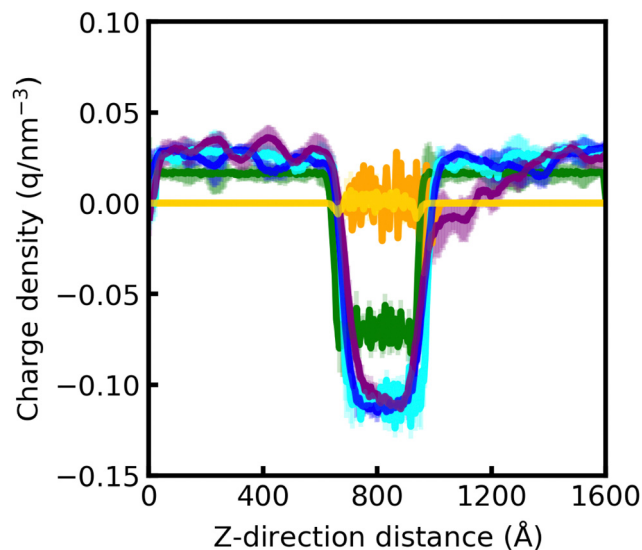


Fig. 6 Charge density profiles of the six systems: pure $E_{25}K_{25}$ (orange), $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1, green), $E_{25}K_{25}/E_{35}K_{15}$ (0.8/0.2, cyan), $E_{25}K_{25}/E_{35}K_{15}$ (0.7/0.3, blue), $E_{25}K_{25}/E_{35}K_{15}$ (0.6/0.4, purple), and $E_{35}K_{15}/E_{15}K_{35}$ (0.5/0.5, gold). Note that the sign of the charge density is positive or negative net-charge.

tensions γ^* are shown in Fig. 7a. They decrease significantly as the molar fraction of $E_{15}K_{35}$ increases to 0.2 and remain stable at the molar fractions of 0.3 and 0.4. Interestingly, the surface tension of the $E_{25}K_{25}/E_{35}K_{15}$ (0.9/0.1) is more than 20% smaller than that of the pure $E_{25}K_{25}$ condensate, even though its condensate density is only 4% less than that of the pure $E_{25}K_{25}$ condensate as seen in Section 3.1. Similarly, the surface tension of the $E_{35}K_{15}/E_{15}K_{35}$ (0.5/0.5) condensate is around 40% smaller than that of the pure $E_{25}K_{25}$ condensate; however, their condensate densities are within 2% difference. The addition of $E_{35}K_{15}$ enhances the dynamics of the polymers in the condensed phases. Fig. 7b and c shows the normalized self-diffusion coefficient ($D^* = D/D_0$, D_0 is the self-diffusion coefficient of $E_{25}K_{25}$ polymers in the pure $E_{25}K_{25}$ condensate) of $E_{25}K_{25}$ and $E_{35}K_{15}$ in the condensed phases. It is obvious that the normalized self-diffusion coefficients of $E_{25}K_{25}$ increase as the initial molar fraction of $E_{35}K_{15}$ increases. This is opposite to the decreasing trend in the surface tension, which suggests that dynamical properties and surface tension are negatively correlated. This negative correlation is also observed in single-species condensates of polyampholytes.²⁸ Besides $E_{25}K_{25}$, the diffusion of $E_{35}K_{15}$ also increases as the molar fraction of $E_{35}K_{15}$ increases to 0.2 and then remains stable in the $E_{25}K_{25}/E_{35}K_{15}$ condensate. In comparison with the $E_{25}K_{25}/E_{35}K_{15}$ condensate, the $E_{35}K_{15}$ polymers diffuse slowly in the $E_{35}K_{15}/E_{15}K_{35}$ condensate (0.5/0.5). This can be attributed to the strong attraction between oppositely charged $E_{35}K_{15}$ and $E_{15}K_{35}$ polymers.

4. Conclusions

We investigate the charge-asymmetric polyampholytes, $E_{35}K_{15}$ and $E_{15}K_{35}$, along with the charge-symmetric $E_{25}K_{25}$. The phase separation of the mixtures of $E_{25}K_{25}/E_{35}K_{15}$ is determined by

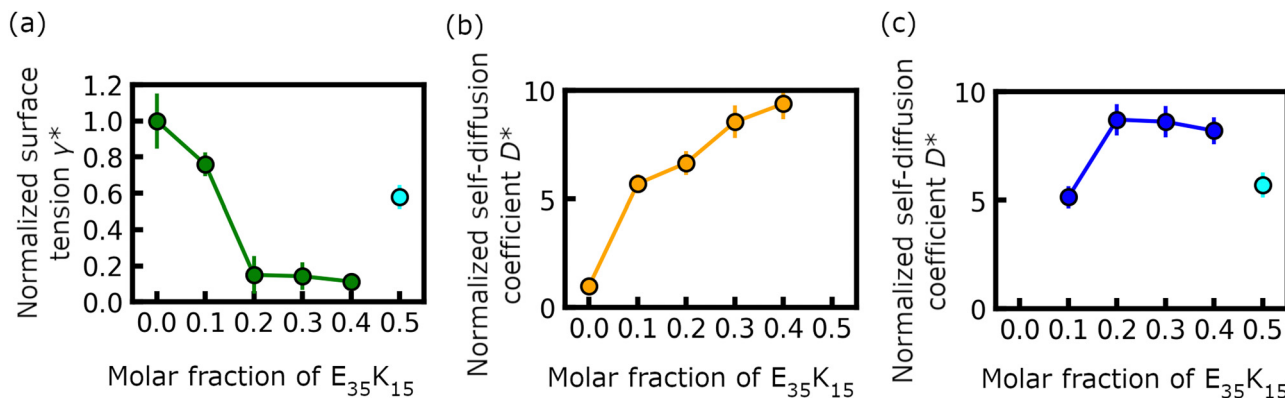


Fig. 7 (a) Normalized surface tensions (γ^*) of the condensates formed by pure E₂₅K₂₅ and mixtures of E₂₅K₂₅/E₃₅K₁₅ and E₃₅K₁₅/E₁₅K₃₅. (b) Normalized self-diffusion coefficients (D^*) of E₂₅K₂₅ in the condensed phases. (c) Normalized self-diffusion coefficients (D^*) of E₃₅K₁₅ in the condensed phases. The cyan data point is for the E₃₅K₁₅/E₁₅K₃₅ (0.5/0.5) condensate.

coarse-grained simulations. The phase-separation propensity is suppressed as the molar fraction of E₃₅K₁₅ increases. This can be attributed to the electrostatic repulsion between E₃₅K₁₅ polymers. Due to the repulsion, the condensate density decreases and eventually no LLPS occurs when the molar fraction of E₃₅K₁₅ increases to 0.5. Meanwhile, the structure and properties of the condensed phases are also affected by the molar fraction of E₃₅K₁₅ polymers. The E₃₅K₁₅ polymers show a tendency to collapse as the molar fraction of E₃₅K₁₅ goes up. The surface tension reduces but the diffusion of polymers increases. Different from the charge-neutral condensates formed by only E₂₅K₂₅ or the mixtures of E₃₅K₁₅/E₁₅K₃₅ with equal molar fractions, the condensates of the mixtures of E₂₅K₂₅/E₃₅K₁₅ (0.9/0.1–0.6/0.4) are negatively charged. The charged condensates offer significant potential for biomedical applications due to their ability to bind with other biomolecules through electrostatic interactions. This property can be harnessed for various purposes such as drug delivery, tissue engineering, and diagnostic assays. Overall, this work provides valuable insights into the phase separation of multi-component polyampholytes with symmetric and asymmetric charge patterning, as well as guidelines for tuning the macroscopic properties of condensates. In the future, we expect the development of coarse-grained models accurately representing the polyampholyte-counterion interactions and will dive into investigating the effects of counterions (e.g. concentrations and valencies) on the LLPS of polyampholytes.

Author contributions

An, Y. conceived and designed the analysis, collected data and performed the analysis, and wrote the paper. Gao, T. performed the simulations and collected the data. Wang, T., Zhang, D., and Bharti, B. reviewed and edited the paper.

Data availability

Data for this article, including all the original data for Fig. 1–7 and Fig. S1, S2 (ESI[†]) are available at figshare at https://figshare.com/articles/figure/Polyampholytes_soft_matter/26047654.

Conflicts of interest

There are no conflicts to declare.

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